

Cracking the Shell

of .NET MAUI

David Ortinau

Principal Product Manager
.NET MAUI

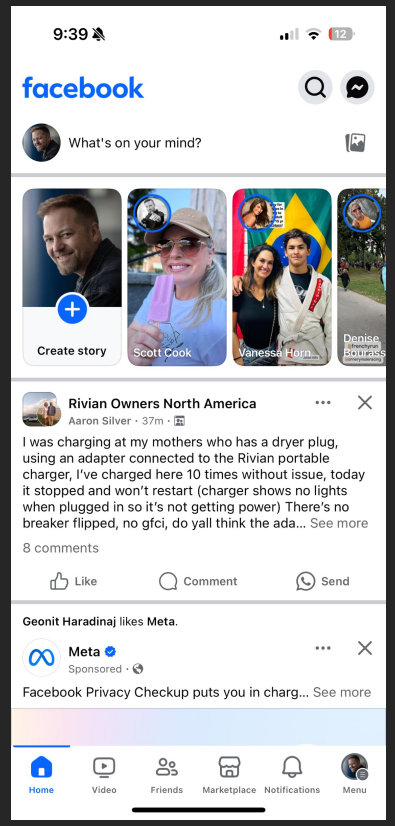
@davidortinau

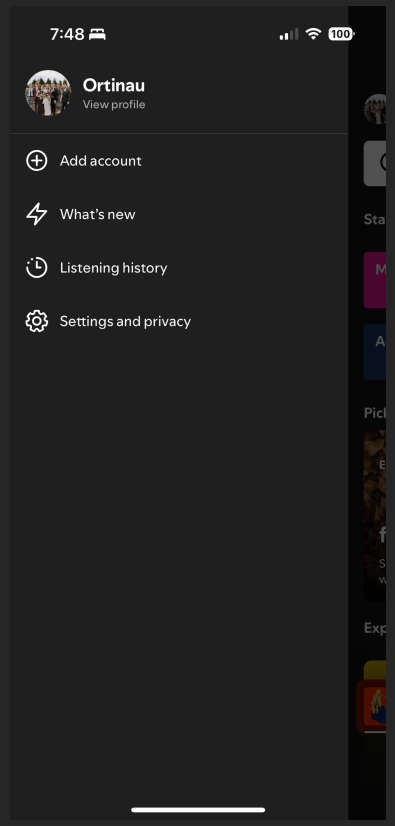
Review my session

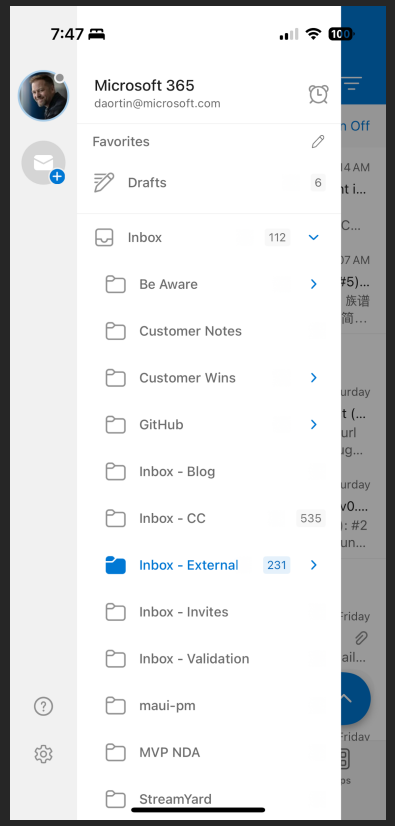


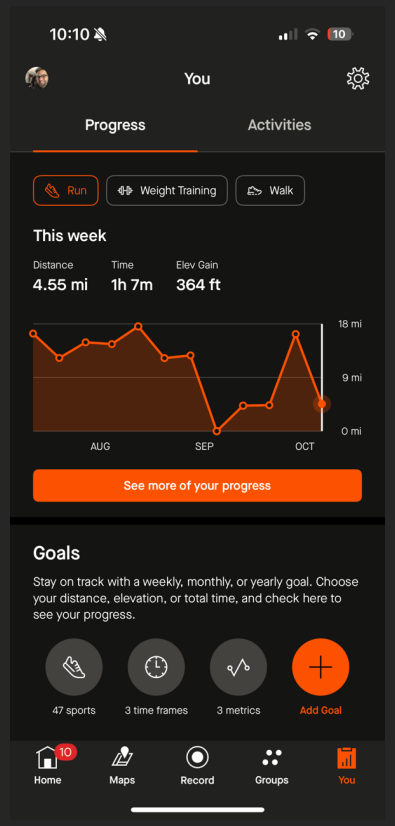
What is Shell

Shell provides a default implementation of common things most all apps need.





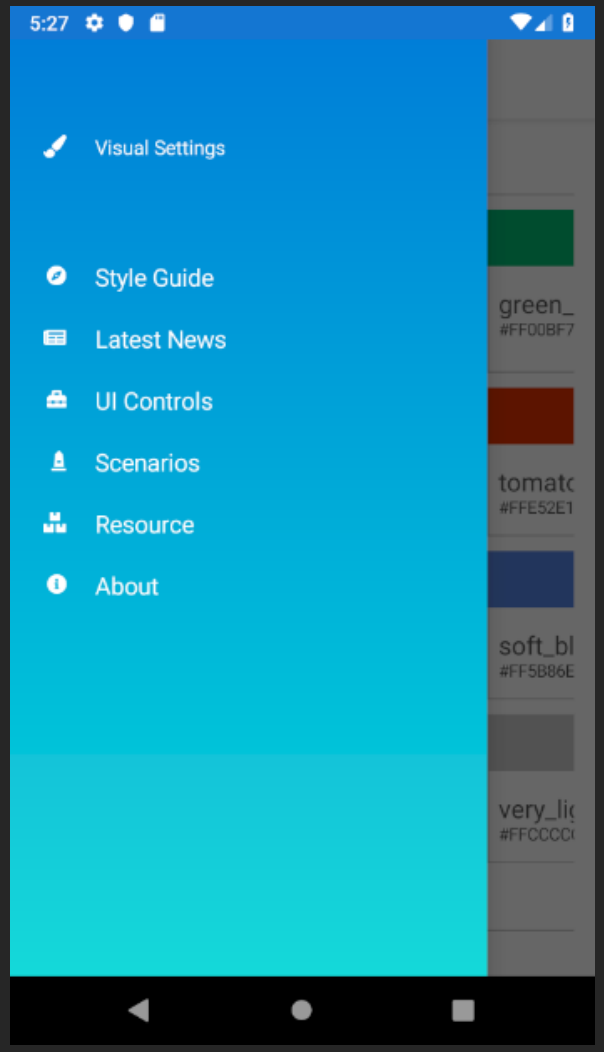


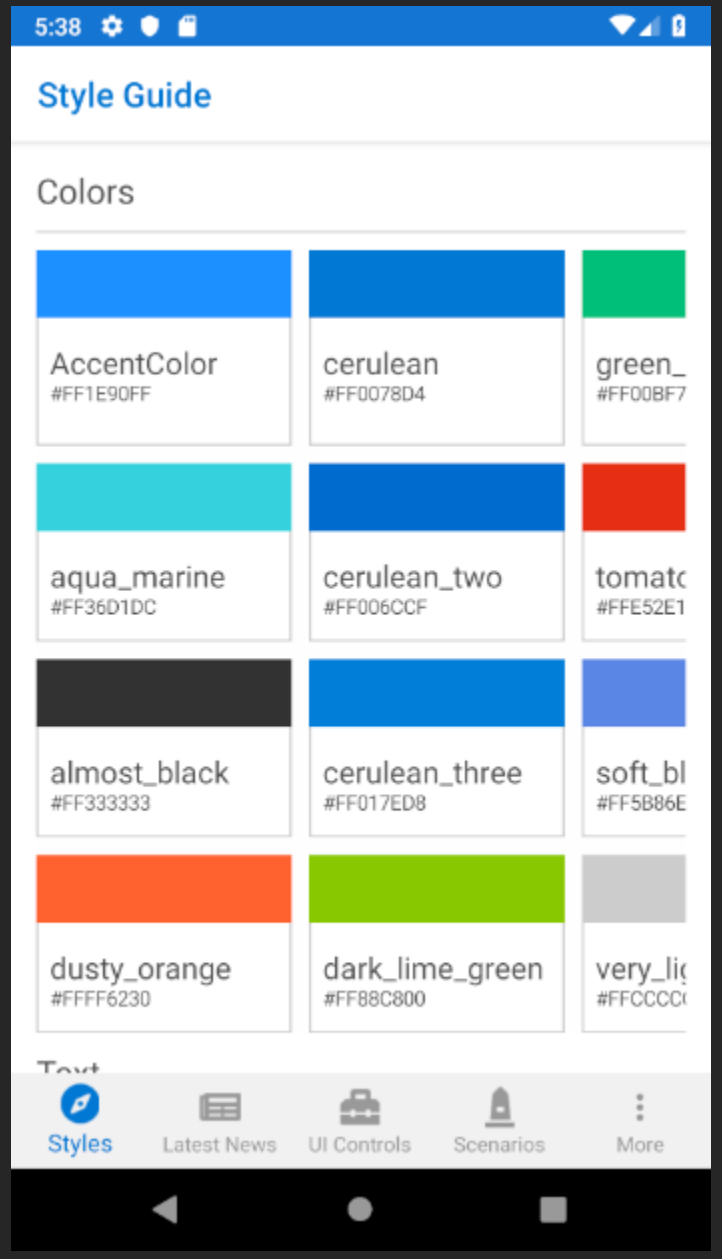


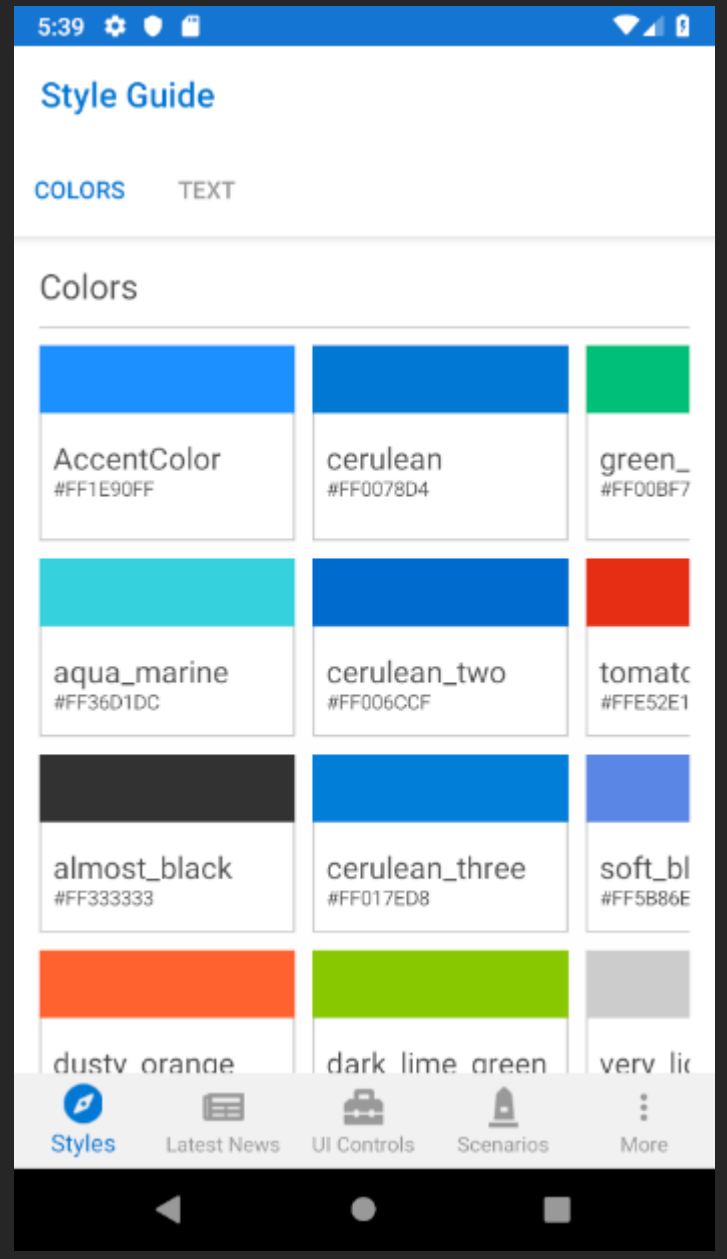
What is Shell

Shell provides a default implementation of common things most all apps need.

- Flyout – aka side drawer
- Tabs – bottom and top
- URI based navigation service
- Back button UX
- Search UX


5:27
Visual Settings
Style Guide
Latest News
UI Controls
Scenarios
Resource
About


5:38
Style Guide
Colors
AccentColor
#FF1E90FF
cerulean
#FF0078D4
aqua_marine
#FF36D1DC
cerulean_two
#FF006CCF
almost_black
#FF333333
cerulean_three
#FF017ED8
dusty_orange
#FFFF6230
dark_lime_green
#FF88C800
Styles
Latest News
UI Controls
Scenarios
More

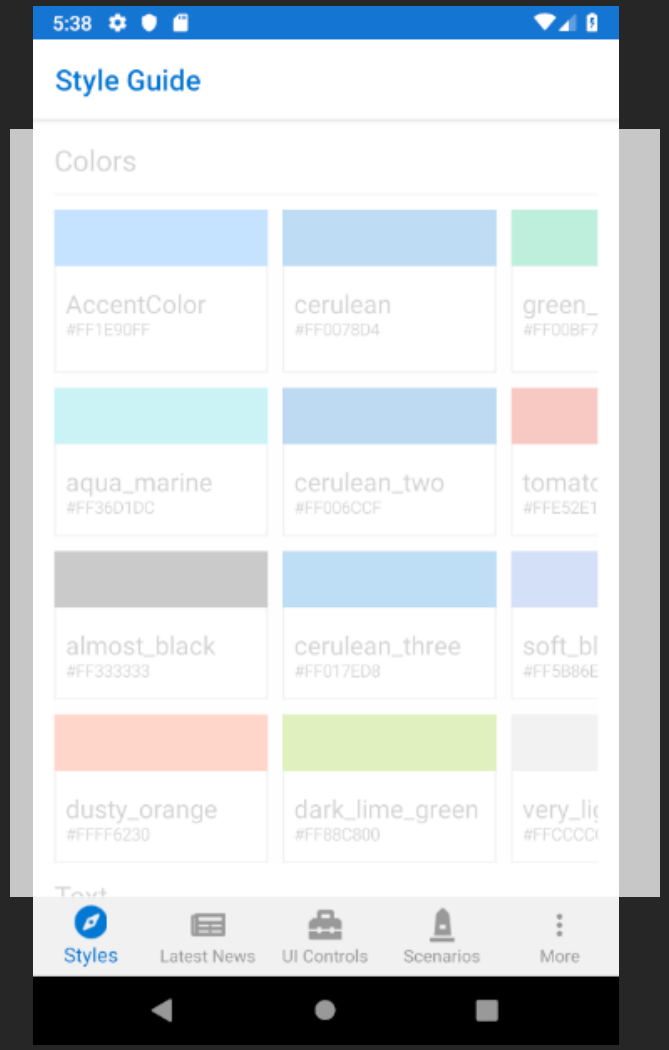

5:39
Style Guide
COLORS
TEXT
Colors
AccentColor
#FF1E90FF
cerulean
#FF0078D4
aqua_marine
#FF36D1DC
cerulean_two
#FF006CCF
almost_black
#FF333333
cerulean_three
#FF017ED8
Styles
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Scenarios
More

Shell

Tabs Made Simple

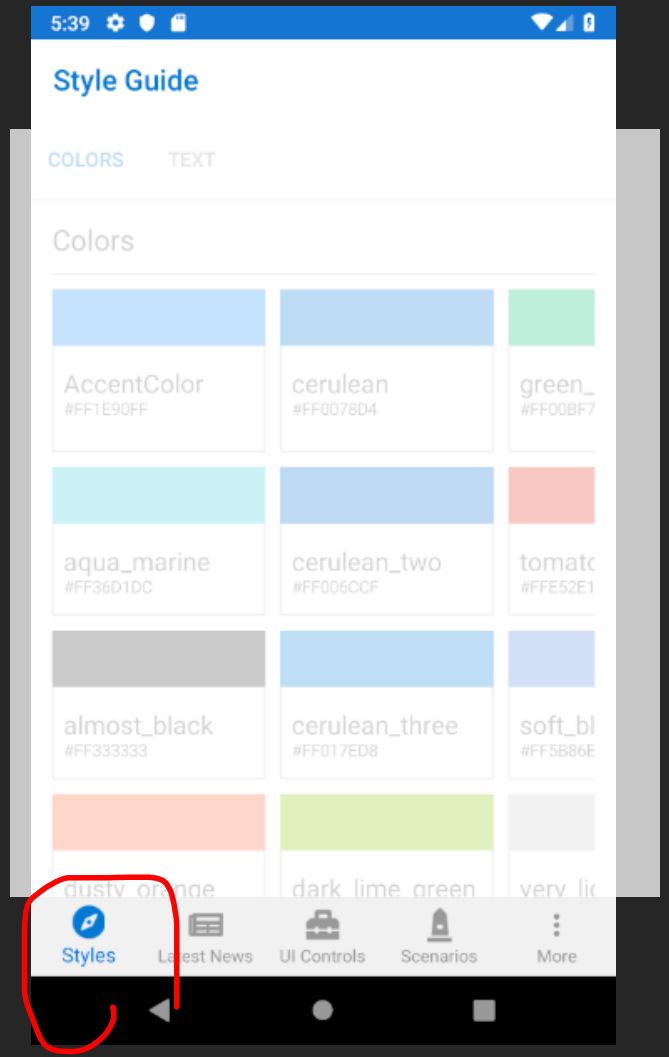
```
<TabBar>
  <Tab Title="Browse" Icon="tab_feed.png">
    <ShellContent ContentTemplate="{DataTemplate local:ItemsPage}" />
  </Tab>
  <Tab Title="About" Icon="tab_about.png">
    <ShellContent ContentTemplate="{DataTemplate local:AboutPage}" />
  </Tab>
</TabBar>
```

```
<TabbedPage x:Class="App5.Views.MainPage">
  <TabbedPage.Children>
    <NavigationPage Title="Browse">
      <NavigationPage.Icon>
        <OnPlatform x:TypeArguments="FileImageSource">
          <On Platform="iOS" Value="tab_feed.png"/>
        </OnPlatform>
      </NavigationPage.Icon>
      <x:Arguments>
        <views:ItemsPage />
      </x:Arguments>
    </NavigationPage>
    <NavigationPage Title="About">
      <NavigationPage.Icon>
        <OnPlatform x:TypeArguments="FileImageSource">
          <On Platform="iOS" Value="tab_about.png"/>
        </OnPlatform>
      </NavigationPage.Icon>
      <x:Arguments>
        <views:AboutPage />
      </x:Arguments>
    </NavigationPage>
  </TabbedPage.Children>
</TabbedPage>
```




```
<Shell>
  <TabBar>
    <Tab
      Title="Styles"
      Icon="Compass.png">
      <ShellContent
        ContentTemplate="{DataTemplate p:StyleGuidePage}"/>
    </Tab>
  </TabBar>
</Shell>
```

Shell Bottom Tabs




```
<Shell>
  <TabBar>
    <Tab
      Title="Style Guide"
      Icon="Compass.png">
      <ShellContent
        Title="Colors"
        ContentTemplate="{DataTemplate p:ColorsPage}"/>
      <ShellContent
        Title="Text"
        ContentTemplate="{DataTemplate p:TextPage}"/>
    </Tab>
  </TabBar>
</Shell>
```

Shell Top Tabs

11:53 LTE 59

Home

12

Your Weekly Snapshot

See More

Activities	Time	Distance
2	1h 30m	4.89 mi
▲ 2	▲ 1h 30m	▲ 4.89

David Ortinau

Today at 7:29 AM · Utrecht, Netherlands

Morning Run

Distance	Elev Gain	Time
2.10 mi	264 ft	28m 50s



1 gave kudos

Home 12 | Maps | Record | Groups | You

11:53 LTE 59

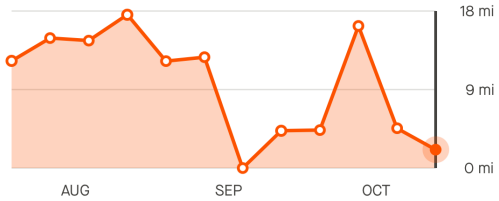
You

Progress | Activities

Run | Weight Training | Walk

This week

Distance	Time	Elev Gain
2.10 mi	28m 50s	264 ft



See more of your progress

Goals

Stay on track with a weekly, monthly, or yearly goal. Choose your distance, elevation, or total time, and check here to see your progress.

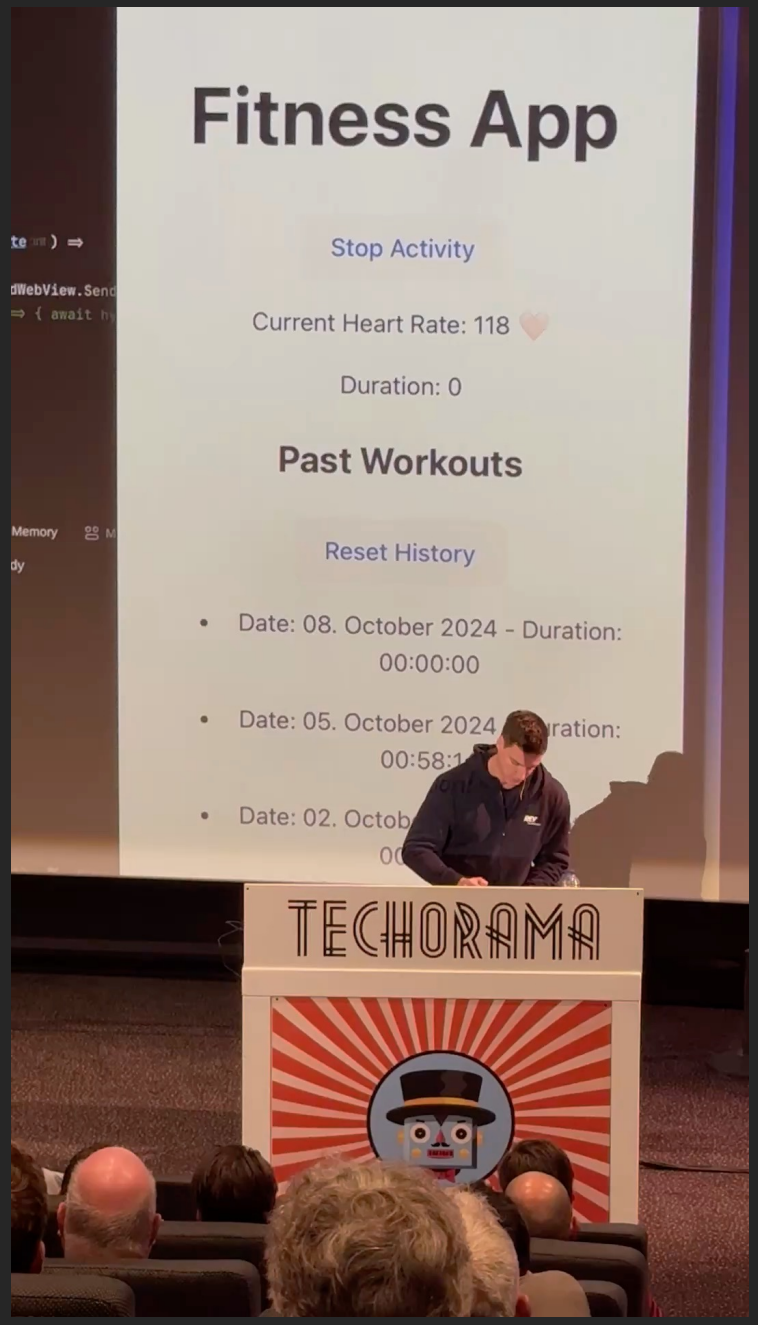
47 sports | 3 time frames | 3 metrics | Add Goal

Home 12 | Maps | Record | Groups | You

Microsoft

Show Me Code!

.NET Health



Flyout Made Simple

```
<FlyoutPage xmlns="http://schemas.microsoft.com/dc
            xmlns:x="http://schemas.microsoft.com/
            xmlns:views=
            x:Class="App
    <FlyoutPage.Flyout>
        <views:MenuPage
    </FlyoutPage.Flyout>
    <FlyoutPage.Detail>
        <NavigationPage>
            <NavigationP
                <OnPlatf
                    <On
                </OnPlat
            </Navigation
            <x:Arguments
                <views:I
            </x:Argument
        </NavigationPage
    </MasterDetailPage.Detail>

</FlyoutPage>
```

```
menuItems = new List<Hom
            {
                new Home
                new Home
            };

            ListViewMenu

            ListViewMenu
            ListViewMenu
            {
                if (e.Se
                    retu

                var id =
                await Ro
            };
```

```
public partial class MainPage : FlyoutPage
    {
        Dictionary<int, NavigationPage> MenuPages = new Dictionary<int, NavigationPage>();
        public MainPage()
        {
            InitializeComponent();

            FlyoutBehavior = FlyoutBehavior.Popover;

            MenuPages.Add((int)MenuItemType.Browse, (NavigationPage)Detail);
        }

        public async Task NavigateFromMenu(int id)
        {
            if (!MenuPages.ContainsKey(id))
            {
                switch (id)
                {
                    case (int)MenuItemType.Browse:
                        MenuPages.Add(id, new NavigationPage(new ItemsPage()));
                        break;
                    case (int)MenuItemType.About:
                        MenuPages.Add(id, new NavigationPage(new AboutPage()));
                        break;
                }
            }

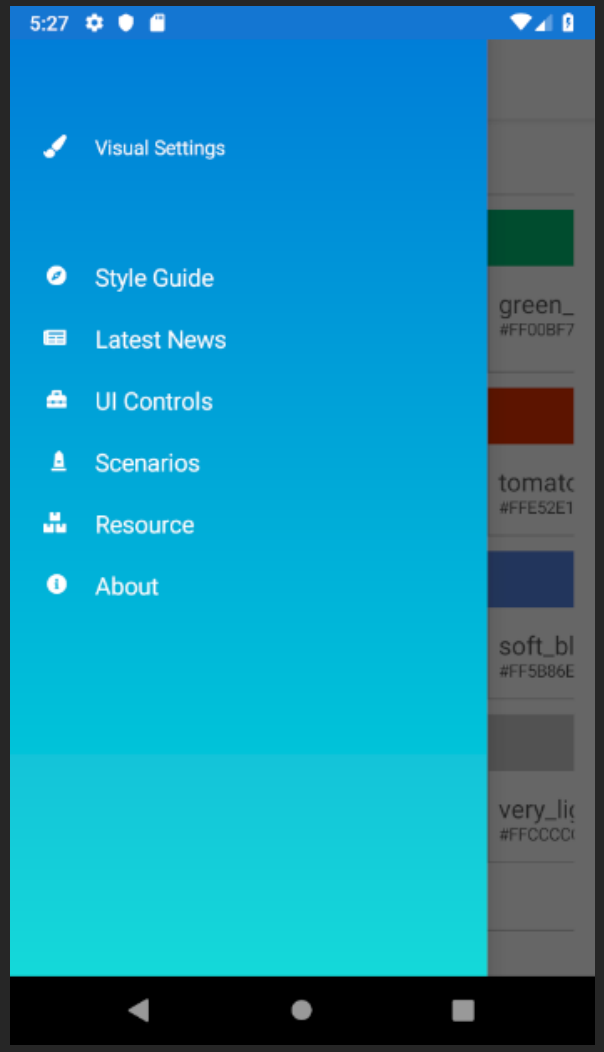
            var newPage = MenuPages[id];

            if (newPage != null && Detail != newPage)
            {
                Detail = newPage;

                if (Device.RuntimePlatform == Device.Android)
                    await Task.Delay(100);

                IsPresented = false;
            }
        }
    }
```

```
</ContentPage>
```




```
<Shell>
  <FlyoutItem
    Title="Style Guide"
    Icon="Compass.png">
    <ShellContent
      ContentTemplate="{DataTemplate p:StyleGuidePage}"/>
  </FlyoutItem>
</Shell>
```

Shell Flyout

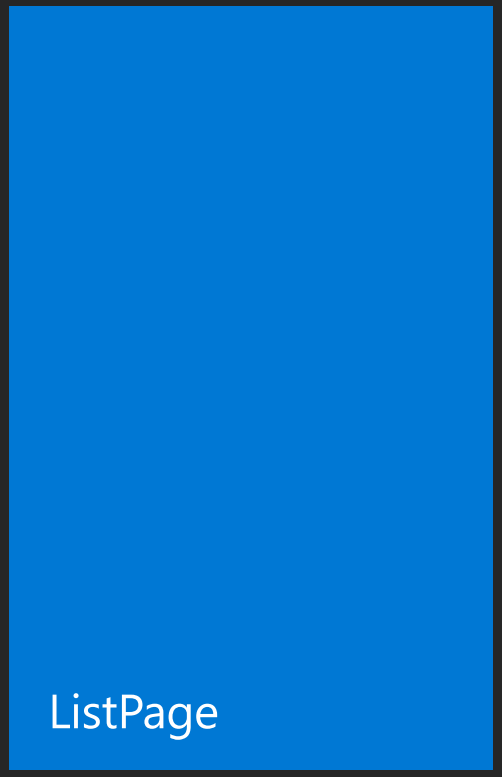
Microsoft

Show Me Code!

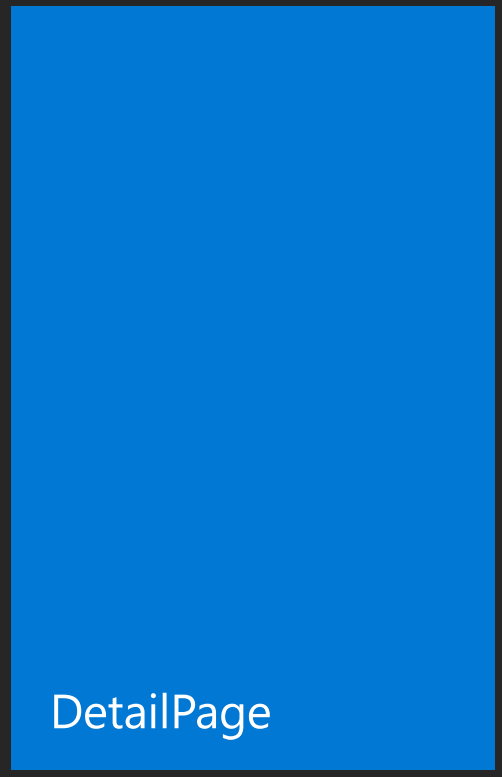
Fly Me

Control Gallery

Point of Sale




```
await Shell.Current
    .GoToAsync($"editVocabulary?id={listID}");
```



Shell Navigation

```
[QueryProperty(nameof(ListId), "id")]
public partial class EditVocabularyPageModel : ObservableObject
{
  [ObservableProperty] private int _listId;

  partial void OnListIdChanged(int value)
  {
    if (value > 0)
    {
    TaskMonitor.Create(LoadList);
    }
  }
}
```

Shell Navigation

```
var payload = new ShellNavigationQueryParameters {
    {"listID", VocabList.ID},
    {"skillProfileID", SkillProfile.ID}
};

try{
    await Shell.Current.GoToAsync($"{route}", payload);
}catch(Exception ex){
    Debug.WriteLine($"{ex.Message}");
}
```

Shell Navigation

```
public partial class ClozurePageModel : IQueryAttributable {
  public void ApplyQueryAttributes(IDictionary<string, object> query){
    ListID = (int)query["listID"];
    SkillProfileID = query.ContainsKey("skillProfileID") ? (int)query["skillProfileID"] : 1;

    // good place to do next thing
  }
}
```

Shell Navigation

```
protected override void OnNavigating(ShellNavigatingEventArgs args)
{
    base.OnNavigating(args);

    // Cancel any back navigation.
    if (args.Source == ShellNavigationSource.Pop)
    {
      args.Cancel();
    }
}
```

Navigation Events

```
public MyShell : Shell
{
    // ...
    protected override async void OnNavigating(ShellNavigatingEventArgs args){
      base.OnNavigating(args);

      ShellNavigatingDeferral token = args.GetDeferral();

      var result = await DisplayActionSheet("Navigate?", "Cancel", "Yes", "No");
      if (result != "Yes"){
        args.Cancel();
      }
      token.Complete();
    }
}
```

Navigation Deferral

Basic Styling

```
<Style TargetType="Label” Class="FlyoutItemLabelStyle">
    <Setter Property="TextColor” Value="Black" />
    <Setter Property="HeightRequest” Value="100" />
</Style>

<Style TargetType="Image” Class="FlyoutItemImageStyle">
    <Setter Property="Aspect” Value="Fill" />
</Style>

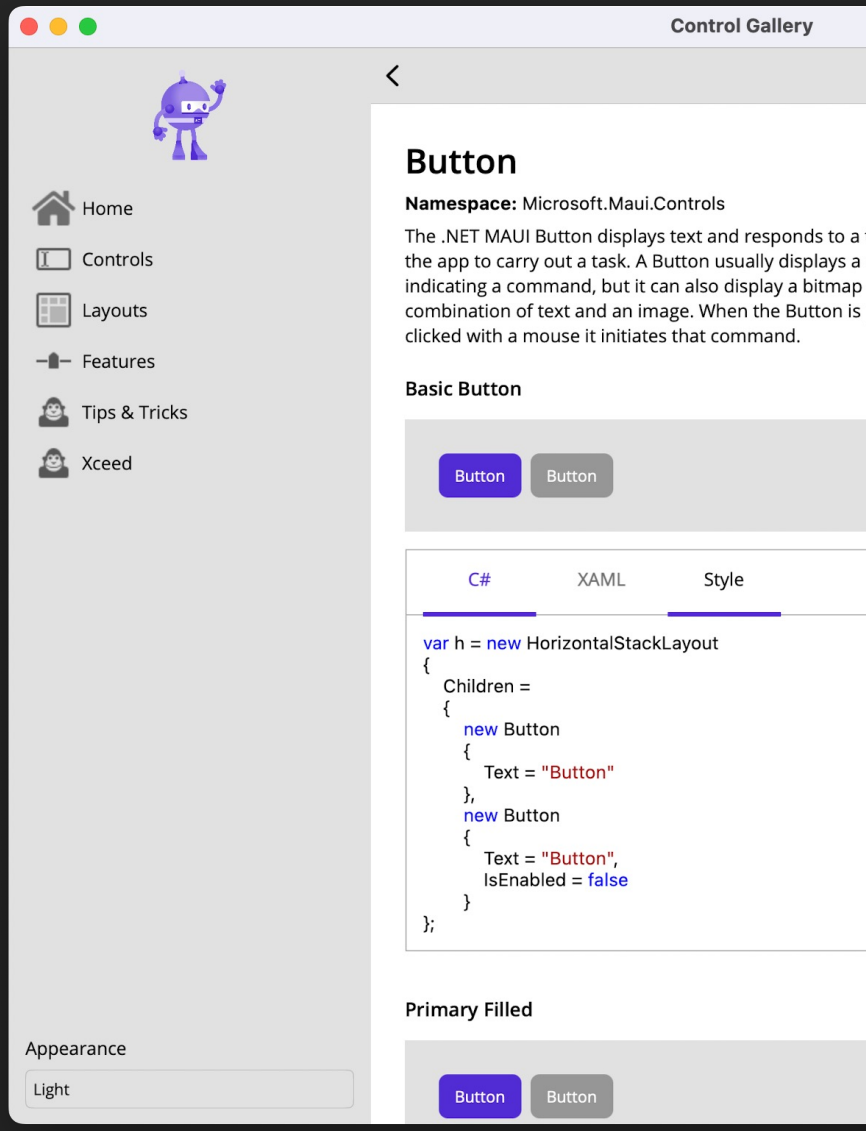
<Style TargetType="Layout” Class="FlyoutItemLayoutStyle” ApplyToDerivedTypes="True">
    <Setter Property="BackgroundColor” Value="Teal" />
</Style>
```

Basic Styling

```
<Style TargetType="Shell" ApplyToDerivedTypes="True">
  <Setter Property="Shell.BackgroundColor" Value="" />
  <Setter Property="Shell.ForegroundColor" Value="" />
  <Setter Property="Shell.TitleColor" Value="" />
  <Setter Property="Shell.DisabledColor" Value="" />
  <Setter Property="Shell.UnselectedColor" Value="" />
  <Setter Property="Shell.NavBarHasShadow" Value="False" />
  <Setter Property="Shell.TabBarBackgroundColor" Value="" />
  <Setter Property="Shell.TabBarForegroundColor" Value="" />
  <Setter Property="Shell.TabBarTitleColor" Value="" />
  <Setter Property="Shell.TabBarUnselectedColor" Value="" />
</Style>
```

Advanced Styling

- Shell.ItemTemplate
- Shell.FlyoutFooter
- Shell.FlyoutHeader
- Shell.FlyoutContentTemplate
- Shell.TitleView



Back Button

```
<ContentPage ...>
  <Shell.BackButtonBehavior>
    <BackButtonBehavior
      Command="{Binding BackCommand}"
      IconOverride="back.png" />
  </Shell.BackButtonBehavior>
  ...
</ContentPage>
```

GoToAsync Routing

```
await Shell.Current.GoToAsync(“..?refresh=true”);
```

Format	Description
route	The route hierarchy will be searched for the specified route, upwards from the current position. The matching page will be pushed to the navigation stack.
/route	The route hierarchy will be searched from the specified route, downwards from the current position. The matching page will be pushed to the navigation stack.
//route	The route hierarchy will be searched for the specified route, upwards from the current position. The matching page will replace the navigation stack.
///route	The route hierarchy will be searched for the specified route, downwards from the current position. The matching page will replace the navigation stack.

Microsoft

Show Me Code!

Time permitting…

David tends to talk a lot

Kinda surprised you made it this far

Thanks! Let’s talk.

@davidortinau

